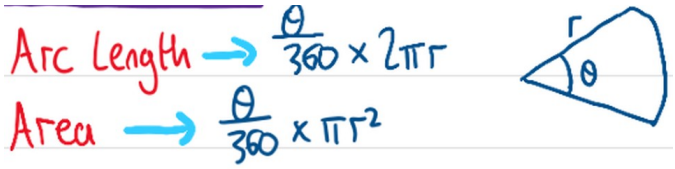
Worksheet 40 · Sectors IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Find the arc length of a sector • Find the area of a sector • Solve problems involving segments and compound shapes		
Grade: / 48	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes



Questions *Sectors*

Question 1 (2 marks)

The diagram shows sector OPQ of a circle, centre O

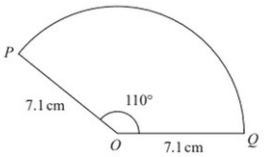


Diagram **NOT** accurately drawn

$OP = OQ = 7.1$ cm
Angle $POQ = 110°$

Calculate the area of sector OPQ
Give your answer correct to one decimal place.

..... cm²

Question 2 (4 marks)

A circle centre O has radius 9 cm.

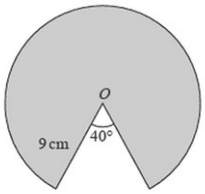


Diagram **NOT** accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

Question 3 (3 marks)

The diagram shows a sector AOB of a circle with centre O

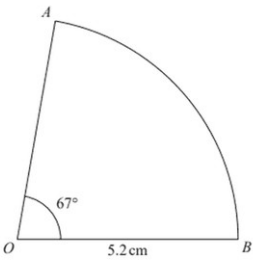


Diagram **NOT** accurately drawn

Angle $AOB = 67°$
$OA = OB = 5.2$ cm

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

Question 4 (4 marks)

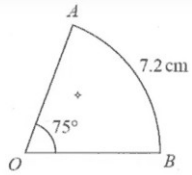


Diagram **NOT** accurately drawn

The diagram shows a sector OAB of a circle, centre O.

Angle $AOB = 75°$
Length of arc $AB = 7.2$ cm

Calculate the area of the sector.
Give your answer correct to 3 significant figures.

Question 5 (6 marks)

The shape $OABC$ is made from a triangle and a sector of a circle.

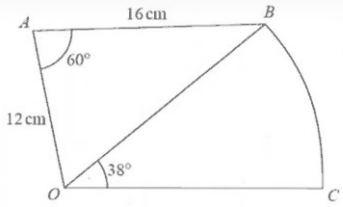


Diagram **NOT** accurately drawn

OAB is a triangle.
OBC is a sector of a circle, centre O.

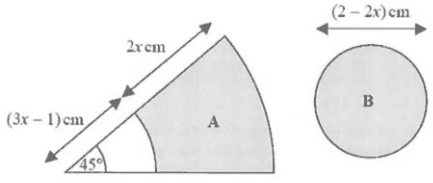
$OA = 12$ cm
$AB = 16$ cm
Angle $OAB = 60°$
Angle $BOC = 38°$

Work out the area of $OABC$.
Give your answer correct to 3 significant figures.

Question 6 (5 marks)

The diagram shows two shaded shapes, **A** and **B**.

Shape **A** is formed by removing a sector of a circle with radius $(3x - 1)$ cm from a sector of the circle with radius $(5x - 1)$ cm.
Shape **B** is a circle of diameter $(2 - 2x)$ cm.



The area of shape **A** is equal to the area of shape **B**.

Find the value of x.
You must show all your working.

Question 7 (2 marks)

The diagram shows a sector of a circle with radius 7 cm.

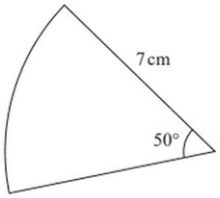
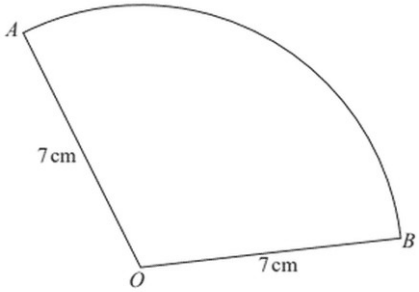


Diagram **NOT** accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

Question 8 (4 marks)

OAB is a sector of a circle with centre O and radius 7 cm.



The area of the sector is 40 cm^2

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

Question 9 (3 marks)

The diagram shows a circle with centre O

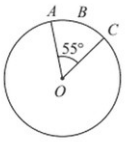


Diagram **NOT** accurately drawn

A, B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55°$

work out the area of the circle.
Give your answer correct to one decimal place.

Question 10 (5 marks)

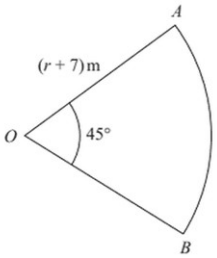


Diagram **NOT** accurately drawn

OAB is a sector **S** of a circle with centre *O* and radius $(r + 7)$ metres.
Angle $AOB = 45°$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r
Show your working clearly.

Question 11 (5 marks)

A, *B* and *C* are points on a circle with centre *O*.

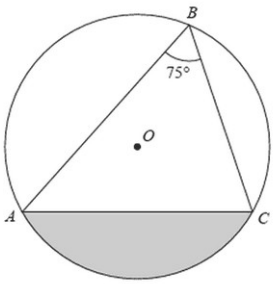


Diagram **NOT** accurately drawn

Angle $ABC = 75°$

The area of the shaded segment is $200\,cm^2$

Calculate the radius of the circle.
Give your answer correct to 3 significant figures.

Question 12 (5 marks)

The diagram shows a sector *OBC* of a circle with centre *O* and radius $(6 + x)$ cm.

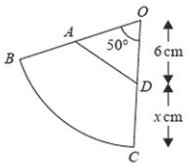


Diagram **NOT** accurately drawn

A is the point on *OB* and *D* is the point on *OC* such that $OA = OD = 6\,cm$

Angle $BOC = 50°$

Given that

the perimeter of sector *OBC* = 2 × the perimeter of triangle *OAD*

find the value of x.
Give your answer correct to 3 significant figures.